

The Human Building Synergy Laboratory

Goal

The Human Building Synergy (HUBS) Laboratory, located in Room 116 of the Civil Engineering Building at the University of Arizona (1209 E. 2nd St., Tucson, AZ 85719), advances the science and technology of the built environment by fostering interdisciplinary research at the intersection of architectural, civil, electrical, and mechanical engineering, and architecture.

The laboratory is purpose-built to investigate how building systems and human occupants interact. Its primary research areas include AI-based occupant comfort control frameworks for smart buildings, indoor environmental quality (IEQ) assessment and improvement, HVAC system performance evaluation under controlled conditions, and human thermoregulation studies across a range of indoor thermal environments. The facility also functions as a living lab, where graduate students occupy the space and serve as research participants, generating continuous, long-term baseline data on occupant comfort and behavior. When not in active experimental use, the facility is available for other faculty members to conduct their own experiments.

[Insert Figure Here]

Suggested content: Photograph of the Civil Engineering Building showing the entrance and building context on campus.

Figure 1. Exterior view of the Civil Engineering Building, University of Arizona.

Facility Layout

The HUBS Laboratory is divided into three distinct zones, each designed to serve a specific role in the research workflow. The layout enables researchers to run parallel experiments under different environmental conditions or to combine spaces for larger-scale studies. All three zones share a uniform finished ceiling height of 8'10".

Chamber 1 is the smaller of the two experimental chambers, measuring 8' wide × 19'10" long. It is equipped with a sliding window for visual observation and emergency egress, as well as a folding door on the shared wall with Chamber 2. This chamber is well suited for tightly controlled experiments where a compact volume helps achieve rapid thermal equilibrium.

Chamber 2 is the larger experimental space, measuring 12'6" wide × 16'6" long. Its greater floor area accommodates more occupants and larger experimental setups. When the folding partition door between the two chambers is opened, Chambers 1 and 2 merge into a single continuous experimental volume, allowing researchers to study airflow patterns, thermal gradients, and occupant behavior across a larger open-plan space.

The Control Room is a separate monitoring and operations space. It houses the data acquisition server and workstations, and provides researchers with a thermally isolated vantage point from which to observe and manage experiments in the adjacent chambers without influencing the experimental conditions.

[Insert Figure Here]

Suggested content: Annotated architectural floor plan showing Chamber 1, Chamber 2, Control Room, folding partition door, sliding window, and entry points. Source: construction documents in _ConstructionDocument/Drawing/Architectural/.

Figure 2. Floor plan of the HUBS Laboratory (Room CE 116).

[Insert Figure Here]

Suggested content: Wide-angle photograph taken from the Control Room looking into the two chambers, showing the suspended ceiling, lighting fixtures, sensor locations, and folding partition door. Source: _Photos/ folder.

Figure 3. Interior photograph of the laboratory.

Capability

Building Automation System (BAS)

The laboratory's building automation system is built on the Delta Controls platform, managed through enteliWEB — a cloud-based software suite that provides unified control of HVAC and lighting systems, real-time monitoring dashboards, and long-term trend data archiving with analytics capabilities. This platform enables researchers to program complex control sequences, log system behavior at fine time intervals, and remotely access building performance data from any networked device.

Each of the three zones is equipped with a Delta eZNT-T304 fully programmable thermostat. Unlike conventional thermostats, the eZNT accepts external sensor inputs for air temperature, relative humidity, CO₂ concentration, and occupancy, allowing researchers to implement custom control logic that responds to multiple environmental variables simultaneously. The thermostats also support cloud connectivity, enabling remote setpoint adjustment and data retrieval.

In addition, each zone is outfitted with a Delta UNOnext (UNO-9SW) indoor air quality monitor. The UNOnext was selected over alternative sensor hubs because it measures a broader set of parameters in a single device: air temperature, relative humidity, PM2.5, PM10, CO₂, illuminance (lux), total volatile organic compounds (TVOC), formaldehyde (HCHO), and carbon monoxide (CO). This consolidated approach reduces sensor clutter while providing comprehensive IEQ data integrated directly into the BAS.

Communication between controllers and field devices uses the BACnet MS/TP protocol, bridged to BACnet/IP through a router for integration with the campus network. The three zones operate as independent thermostatic control zones, meaning each chamber and the control room can maintain different temperature setpoints and ventilation strategies simultaneously.

[Insert Figure Here]

Suggested content: Schematic showing the Delta Controls platform: eZNT thermostats, UNOnext IAQ monitors, BACnet communication path, and enteliWEB cloud interface. Could be adapted from control system proposal documents in _Contracts/Control/.

Figure 4. BAS system architecture diagram.

HVAC System

The HVAC system is designed to give researchers precise, zone-level control over the thermal environment in each of the three spaces. Each zone is served by a dedicated Trane M-Series horizontal concealed fan coil unit (Size 040, 115 V / 60 Hz / 1 Ph), mounted above the suspended ceiling. The fan coil units feature a 4-row cooling coil with a stainless steel drain pan for chilled water cooling, and a hot water reheat coil, enabling both heating and cooling within the same air-handling path. A high-static fan motor ensures adequate air distribution even through extended duct runs, and each unit is fitted with MERV 13 filtration to maintain high indoor air quality during experiments.

Airflow modulation is managed through deluxe ball valves on the chilled and hot water supply lines, controlled by 2-way analog valves (2–10 VDC) that receive signals from the BAS. Each fan coil unit has its own outdoor air ventilation connection ducted to the FCU return, allowing controlled introduction of fresh air without directly exposing the chamber to unconditioned outdoor conditions.

A key design feature of the lab is its experimental flexibility. The suspended ceiling uses smaller tiles (12" × 12" or 18" × 18", rather than the standard 24" × 24") so that supply diffusers can be repositioned without major ceiling modification. Flexible ductwork connects the fan coil units to the diffusers, making it straightforward to reconfigure air distribution patterns between experiments. In addition to the fixed HVAC system, the lab maintains supplemental portable equipment — including electric heaters (radiator and convection types), portable air conditioning units, and air purifiers — to extend the range of thermal and air quality conditions that can be studied.

[Insert Figure Here]

Suggested content: Annotated diagram or photograph showing the above-ceiling fan coil units, flexible ductwork, diffuser locations, and chilled/hot water piping connections. Could reference HVAC options drawings from _HVAC/ or mechanical plans from _ConstructionDocument/.

Figure 5. HVAC system layout and ductwork configuration.

Environmental & IEQ Sensors

The laboratory is densely instrumented to capture spatial variation in indoor environmental conditions. Rather than relying on a single sensor at one location, multiple sensors of each type are distributed throughout each zone to map gradients in temperature, humidity, air quality, and other parameters across the occupied space. The following sensor deployment is replicated in each of the three zones:

- **Air Temperature** — Five wall-mounted multi-functional sensors are distributed at different heights and locations to capture vertical and horizontal temperature stratification within the space.
- **Relative Humidity** — Five wall-mounted sensors, co-located with the temperature sensors, track moisture levels that influence both occupant comfort and building material performance.
- **Mean Radiant Temperature** — One ceiling-mounted sensor per zone measures the average surface temperature of surrounding walls, floor, and ceiling. This parameter is critical for thermal comfort research, as radiant heat exchange accounts for a substantial portion of perceived warmth or coolness.
- **Carbon Dioxide (CO₂)** — Three sensors per zone (ppm) monitor CO₂ as a proxy for ventilation effectiveness and occupancy density.
- **Particulate Matter (PM_{2.5} and PM₁₀)** — Three sensors each per zone (µg/m³) detect fine and coarse particulate matter relevant to respiratory health and filtration performance research.
- **Total Volatile Organic Compounds (TVOC)** — Three sensors per zone (µg/m³) identify the presence of volatile organic compounds from building materials, furnishings, and occupant activities.
- **Illuminance** — Three sensors per zone (lux) measure light levels for visual comfort and daylighting studies.
- **Sound Level** — Three sensors per zone (dBA) record ambient noise, supporting research on acoustic comfort and its interaction with thermal conditions.

In addition to the distributed sensor arrays, each zone's UNOnext (UNO-9SW) IAQ monitor provides a consolidated, BAS-integrated measurement of temperature, RH, PM_{2.5}, PM₁₀, CO₂, lux, TVOC, HCHO, and CO. A single outdoor ambient sensor is also installed to measure external temperature, humidity, and CO₂, providing the reference baseline needed to evaluate building envelope and ventilation performance.

[Insert Figure Here]

Suggested content: Plan-view diagram showing the spatial distribution of wall-mounted multi-functional sensors, ceiling-mounted MRT sensor, anemometers, and UNOnext IAQ monitor within one chamber. Include sensor heights on a section view if available.

Figure 6. Sensor placement layout.

Occupancy & Presence Sensing

Understanding how occupants use and move through a space is central to the lab's research mission. The facility employs a multi-technology approach to occupancy detection so that researchers can compare sensor modalities, fuse data streams for higher accuracy, and study occupant behavior under varying room conditions.

Each zone in Chambers 1 and 2 is equipped with three passive infrared (PIR) sensors, three ultrasonic sensors, and three Doppler radar sensors. Deploying three distinct sensing technologies side by side enables cross-validation and allows researchers to benchmark each technology's detection accuracy, response latency, and sensitivity to different occupant behaviors and room configurations. For example, PIR sensors excel at detecting gross body movement, while Doppler radar can register subtle motions such as typing or breathing, making the combined array well suited for studying the full spectrum of sedentary and active occupancy patterns.

[Insert Figure Here]

Suggested content: Diagram showing PIR, ultrasonic, and Doppler radar sensor locations within each chamber, annotated with sensing coverage zones.

Figure 7. Occupancy sensor placement in Chambers 1 and 2.

HVAC Monitoring Sensors

Beyond the room-level environmental sensors, a dedicated set of instrumentation monitors the performance of the HVAC system itself. This allows researchers to correlate energy consumption and system behavior with the resulting indoor conditions and occupant responses.

- **Hydronic System Monitoring** — Flow meters and insertion-type temperature sensors on both the chilled water and hot water loops measure supply and return temperatures (in °F or °C) and volumetric flow rates (GPM or L/s). These measurements enable calculation of thermal energy delivered to or removed from each zone.
- **Fan Coil Unit Power Monitoring** — Current transducers installed on each fan coil unit's electrical feed measure real-time electrical demand (kW) and cumulative energy consumption (kWh) on a per-room basis. Separate circuits for each zone make it possible to attribute energy use directly to specific experimental conditions.
- **Duct and Coil Instrumentation** — Duct-mounted and coil-mounted temperature sensors (with thermowell protection) track discharge air temperature and coil performance. Fan relays and current sensors (six total) monitor fan operation status across all three units.
- **Filter Condition Monitoring** — Analog differential pressure transmitters across each filter track filter loading over time, supporting research on filtration effectiveness and its impact on system energy consumption and indoor air quality.

[Insert Figure Here]

Suggested content: Piping and instrumentation diagram (P&ID) showing flow meters, temperature sensors, current transducers, and differential pressure transmitters on the hydronic loops and fan coil units.

Figure 8. HVAC monitoring instrumentation schematic.

Lighting System

The lighting system is designed for both functional illumination and experimental control. The facility uses LED fixtures throughout — 10 in Chamber 1, 12 in Chamber 2, and 8 in the Control Room — chosen for their low heat output, which minimizes interference with thermal comfort experiments. Fixtures can be controlled individually or in small groups of two to four, enabling researchers to create non-uniform lighting distributions for studies on visual comfort, task performance, and occupant preference.

In addition to the overhead LED fixtures, the lab integrates Philips Hue White and Color Ambiance smart bulbs, which provide color-tunable output across a wide correlated color temperature (CCT) range. This capability supports circadian lighting research and studies on the psychological effects of lighting spectrum on occupant well-being. Lighting can be operated manually, through the BAS via enteliWEB, or through networked smart-home protocols, giving researchers flexibility in how lighting scenarios are triggered and sequenced during experiments.

[Insert Figure Here]

Suggested content: Reflected ceiling plan showing LED fixture positions, control groupings, and Philips Hue smart bulb locations. Annotate which fixtures belong to each control group.

Figure 9. Lighting fixture layout.

Server & IT Infrastructure

The volume of data generated by the lab's sensor arrays and BAS requires substantial on-site computing and storage capacity. The facility's IT backbone is anchored by a high-specification server equipped with two GPUs, providing the processing power needed for real-time data ingestion, machine learning model training (e.g., occupant comfort prediction, anomaly detection), and computationally intensive building simulation workloads. The server also provides centralized storage for raw sensor data and processed datasets.

Two 27" all-in-one desktop workstations in the Control Room serve as the primary researcher interfaces for experiment configuration, live monitoring, and data analysis. The facility connects to the campus network via wireless connectivity, and a per-circuit energy monitoring system tracks electrical consumption across all loads in the lab, providing granular energy data that can be correlated with environmental conditions and occupancy patterns.

[Insert Figure Here]

Suggested content: Photograph of the Control Room showing the server, workstations, and monitoring displays. Alternatively, a network topology diagram showing the server, workstations, wireless access point, and BAS integration.

Figure 10. Control Room and IT infrastructure.

Research Modes

The combination of independent zone control, reconfigurable chamber geometry, and dense multi-modal sensing enables the HUBS Laboratory to operate in several distinct experimental modes:

- **Cooling Mode** — Investigates passive cooling strategies, natural ventilation through operable windows, and evaporative cooling effects. Researchers can open the sliding window in Chamber 1 or adjust outdoor air dampers to study how Tucson’s arid climate interacts with different ventilation approaches.
- **Heating Mode** — Evaluates occupant thermal comfort under elevated indoor temperatures using the hot water reheat coils and supplemental portable heaters. This mode is particularly relevant for studying adaptive comfort models and thermostat setpoint optimization in warm climates.
- **Heat Transfer Mode** — Focuses on radiative and convective heat transfer between building surfaces, HVAC system components, and human occupants. Mean radiant temperature sensors and the distributed air temperature array provide the spatial resolution needed to characterize heat transfer pathways.
- **Daylighting Mode** — Studies the integration of natural daylight with artificial lighting and its combined effect on visual comfort, energy use, and occupant circadian response. The color-tunable lighting system and illuminance sensors enable precise characterization of the luminous environment.

Because the two chambers can be operated independently or merged into a single volume through the folding partition door, researchers can run simultaneous comparative experiments (e.g., two different HVAC strategies side by side) or sequential experiments in a unified open-plan configuration. The Control Room remains thermally isolated throughout, ensuring that researcher presence does not influence the experimental environment.